



GILBERT IMMANUEL HASIROLAN

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Jakarta, Indonesia

Gilbert is a fresh IT graduate specializing in Data Analytics, Machine Learning, and AI. He builds end-to-end systems that turn messy, real-world data into reproducible pipelines, predictive models, and decision-ready dashboards, from fine-tuning OCR and LLM models for Indonesian essay scoring to deploying a live football-scouting analytics app. He is drawn to problems where careful data work produces clear, defensible answers.

Education

Institut Teknologi Sepuluh Nopember - Surabaya <i>Bachelor of Information Technology, 3.75/4.00</i> - Asst. Lecturer: Data Structure Course - Thesis: Optimising Content Quality and Providing Contextual Feedback in Indonesian Essays Using Optical Character Recognition (OCR) And Large Language Models (LLM)	Jun 2021 - Aug 2025
Bangkit Academy By Google, GoTo, Traveloka  <i>Machine Learning Cohort</i> - Related Courses: Completed comprehensive curriculum covering ML fundamentals, deep learning, computer vision, NLP, and cloud-based ML solutions. Enhanced professional skills through industry-standard English and leadership training. - Project: FoodWise – Developed ML models for a mobile app reducing food waste via expiry tracking and intelligent recipe suggestions.	Feb 2024 - Jul 2024
SMAN 81 Jakarta - Jakarta, Indonesia <i>High School Diploma in Science Major, 87.05/100.00</i>	Jun 2018 - May 2021

Experience

Avalon AI - Remote <i>Front-End Developer</i> - Led front-end development of Summasphere, an award-winning AI paper-summarizer, lifting page-load and navigation efficiency by 20% using ReactJS, D3.js, and TailwindCSS. - Built interactive D3.js visualizations integrated with ExpressJS APIs for AI summarization and topic modeling, helping researchers judge paper relevance in 20% less time.	Apr 2024 - Jun 2024
A Renewal Agent 5.0 - Surabaya <i>Expert Staff of Olympiad Division</i> - Designed cybersecurity and machine-learning olympiad modules that drew 136 preliminary participants and advanced 10 finalists across the competition. - Authored 30 rigorous, fairness-checked assessment questions that accurately ranked participants' cybersecurity and ML knowledge across preliminary and final rounds. - Partnered with faculty to validate content integrity, enabling smooth end-to-end execution from preliminaries through finals.	Sep 2023 - May 2024
Himpunan Mahasiswa Teknologi Informasi ITS (HMIT ITS) - Surabaya <i>Staff of Research and Technology Department</i> - Led IoT club initiatives that recruited 12 new members and ran structured, hands-on hardware projects, raising active participation in the department. - Built and launched a certifications-and-competitions portal that engaged 260+ students across IT fields, centralizing opportunities in one platform. - Mentored 60 students through the portal to earn certifications and enter competitions, lifting their credential and contest-participation rates.	Mar 2023 - Feb 2024
A Renewal Agent 4.0 - Surabaya <i>Expert Staff of Cyber Security and IoT Olympiad Division</i> - Developed cybersecurity olympiad modules and exam questions for preliminary and final rounds under lecturer guidance, ensuring fair, accurate assessment. - Led development of an online preliminaries platform that ran the competition end-to-end, giving participants user-friendly, centralized access to rounds and submissions. - Attracted 70 teams and 140 participants from across Indonesia, expanding the olympiad's national reach.	Oct 2022 - Apr 2023

Projects

Real Madrid Scouting Analysis

Python, Pandas, Streamlit, Jupyter, Matplotlib

- Distilled 7,000+ player-seasons from Europe's Big-5 leagues into a ranked, role-based shortlist of 641 players by merging eight FBref tables into a cleaned pool scored against a custom tactical model.
- Salvaged the project after its main data source was removed mid-build by re-architecting the pipeline around an archived snapshot read in Python, removing all live-scraping dependencies.
- Built a transparent three-lens scoring engine (pool quality, tactical fit, squad need) and deployed it as a live Streamlit dashboard with per-player percentile radars for non-technical users.

Indonesian Essay AI Scoring System

Python, PyTorch, TrOCR, LLM, OpenCV

- Fine-tuned TrOCR on 30,000 synthetic Indonesian handwriting samples and achieved 87.6% character accuracy, outperforming EasyOCR and Tesseract baselines.
- Built end-to-end ML pipeline integrating text detection, multi-level correction with SymSpell and LLM, and automated essay scoring with contextual feedback.
- Developed functional web application prototype demonstrating full system integration and evaluated LLM quality via RAGAS framework.

FoodWise – Food Waste Reduction App

Python, TensorFlow Lite, Flask, Jupyter Notebook

- Created a mobile application that minimizes household food waste by tracking groceries and offering intelligent recipe and freshness recommendations through machine learning.
- Built a content-based filtering system for personalized recipe recommendations matching user inventory data with curated datasets.
- Developed a food freshness detection model using TensorFlow Lite and integrated the ML API with Flask for seamless system communication.

Summasphere – Research Paper Summarizer App

ReactJS, D3JS, ExpressJS, TailwindCSS

- Created an award-winning web application that helps researchers rapidly extract key insights from academic papers, featuring AI-powered summarization and topic-modelling engines.
- Led front-end development using ReactJS and D3JS, with TailwindCSS styling and ExpressJS backend integration to support interactive visualisations and intuitive dashboards.
- Enabled significant workflow improvements for researchers by allowing quick relevance assessment of literature, reducing time spent on non-essential studies and helping stay current with emerging research.

Skills

- **Programming Languages:** Python | SQL | JavaScript | Bash | PHP | HTML | CSS
- **Frameworks & Libraries:** Pandas | NumPy | PyTorch | TensorFlow | Streamlit | Matplotlib | Flask | React.js | Express.js
- **Dev Tools & Platform:** Git | Jupyter Notebook | PostgreSQL | MySQL | Tableau | Power BI | Microsoft Excel | Figma | Canva
- **Languages** 🌐: Native in Bahasa Indonesia | Full Professional Proficiency in English (IELTS 8.0)